

# Teaching NeuroImages: The Venous System and Developmental Venous Anomalies

## Drivers of Vascular Malformations?

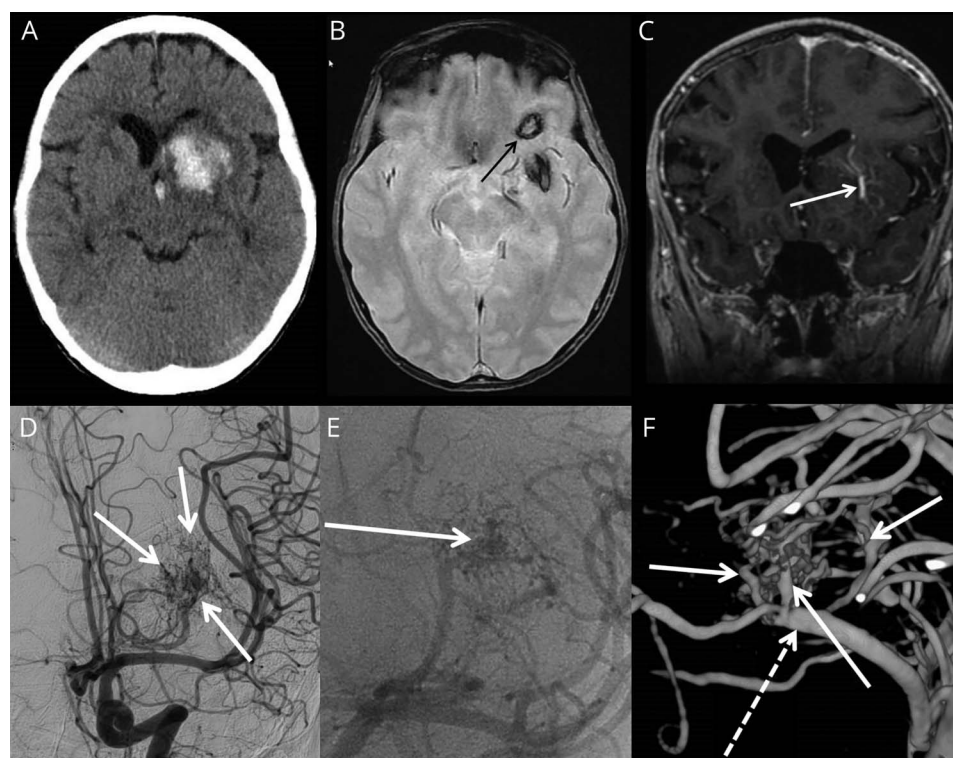
Anthony S. Larson, BS, Harry Cloft, MD, PhD, Ivan D. Carabenciov, MD, and Giuseppe Lanzino, MD

*Neurology*® 2021;96:e960-e961. doi:10.1212/WNL.0000000000010815

### Correspondence

Anthony S. Larson  
lars4689@umn.edu

**Figure** Imaging Features of a CM Concurrent with DVA and AVM



(A) Left-hemispheric hemorrhage. MRI/MRA suggestive of a (B) CM (arrow) and (C) DVA (arrow). (D) A basal ganglia AVM (arrows) with (E) an intranidal aneurysm (arrow), the likely source of hemorrhage. (F) 3-D angiography showing the DVA with smaller veins (solid arrows) and a larger collecting vein (dashed arrow). AVM = arteriovenous malformation; CM = cavernous malformation; DVA = developmental venous anomaly.

A 57-year-old woman presented with acute confusion. CT-scan showed a left-sided hemorrhage, and MR-imaging demonstrated an adjacent cavernous malformation (CM) associated with a complex developmental venous anomaly (DVA) continuous with the hemorrhage (figure, A–C). Digital subtraction angiography demonstrated an arteriovenous malformation (AVM) draining into the DVA (figure, D–F). DVAs are typically benign lesions commonly associated with CMs and rarely, AVMs.<sup>1</sup> Concurrence of these 3 malformations is exceptionally rare. Preserving DVA outflow is crucial when AVM extirpation is indicated.<sup>1</sup> Our understanding of the role of the venous system and DVAs in the genesis of vascular malformations continues to evolve.<sup>2</sup>

### MORE ONLINE

#### →Teaching slides

[links.lww.com/WNL/B214](https://links.lww.com/WNL/B214)

From the Department of Radiology (A.S.L., H.C., G.L.); Department of Neurosurgery (G.L.); and Department of Neurology (I.D.C.), Mayo Clinic, Rochester, MN. Go to [Neurology.org/N](https://Neurology.org/N) for full disclosures. Funding information and disclosures deemed relevant by the authors, if any, are provided at the end of the article.

## Study Funding

No targeted funding reported.

## Disclosure

The authors report no disclosures relevant to the manuscript. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

## Appendix Authors

| Name                         | Location                                            | Contribution                                                                                                     |
|------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Anthony S. Larson, BS</b> | Department of Radiology, Mayo Clinic, Rochester, MN | Design and conceptualized study; analyzed the data; drafted the manuscript for intellectual content              |
| <b>Harry Cloft, MD, PhD</b>  | Department of Radiology, Mayo Clinic, Rochester, MN | Design and conceptualized study; analyzed and acquired the data; drafted the manuscript for intellectual content |

## Appendix (continued)

| Name                           | Location                                                                        | Contribution                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Ivan D. Carabenciov, MD</b> | Department of Neurology, Mayo Clinic, Rochester, MN                             | Design and conceptualized study; analyzed and acquired the data; drafted the manuscript for intellectual content |
| <b>Giuseppe Lanzino, MD</b>    | Department of Radiology; Department of Neurosurgery, Mayo Clinic, Rochester, MN | Design and conceptualized study; analyzed the data; drafted the manuscript for intellectual content              |

## References

1. De Maria L, Lanzino G, Flemming KD, Brinjikji W. Transitional venous anomalies and DVAs draining brain AVMs: a single-institution case series and review of the literature. *J Clin Neurosci* 2019;66:165–177.
2. Aboian MS, Daniels DJ, Rammos SK, Pozzati E, Lanzino G. The putative role of the venous system in the genesis of vascular malformations. *Neurosurg Focus* 2009;27:E9.